Six Failure Modes: A Pre-Deployment Diagnostic Framework for Tokenized Collaborative Economies

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Abstract

Community token economies are gaining traction as tools for coordinating collective action and enabling decentralized value exchange in collaborative, socially-oriented contexts. Yet the formal conditions under which such systems remain sustainable, particularly at the small community scale, are underspecified in existing literature. This paper proposes a diagnostic framework comprising six design-stage failure modes: token oversupply, free-rider collapse, velocity trap, insufficient critical mass, governance capture, and burn/emission imbalance. Each failure mode is associated with a minimum parameter threshold derived from monetary economics, token engineering literature on micro-velocity, and Ostrom's commons governance principles. The framework is deliberately scoped to design-stage parameters, requiring no longitudinal transaction data, making it actionable before deployment when corrective intervention is most feasible. The framework is applied to three real-world community token economy projects at design stage, producing structured risk profiles and a cross-case comparative analysis. The paper concludes by positioning the framework as the analytical and parametric foundation for a forthcoming agent-based model of community token economy dynamics.

Keywords

Token engineering, Token economy, Collaborative economies,

1. Introduction

The emergence of blockchain-based token economies has generated considerable interest as a mechanism for coordinating collective action, incentivizing prosocial behavior, and enabling decentralized value exchange in community contexts (Lesche, Sandner, Treiblmaier, 2022; Kivilo et al., 2026). Unlike cryptocurrency systems designed for speculative or financial purposes, community token economies are explicitly oriented toward social goals: rewarding voluntary participation, facilitating the exchange of goods and services among peers, redistributing resources in informal economy conetxts, or enabling democratic governance of shared assets.

The academic literature on tokenomics has developed primarily in relation to large-scale blockchain protocols where transaction volumes are high, participant bases are large, and market mechanisms can be assumed to operate efficiently (Freni, Ferro, Moncada, 2020; Tan, 2019). The formal conditions under which a token economy achieves sustainability remain poorly specified in the literature, particularly at the community scale. This paper addresses that gap. We identify six failure modes that recurrently threaten the sustainability of community token economies, token oversupply and inflation, free-rider and contribution collapse, token velocity trap, insufficient critical mass, governance capture, and burn/emission imbalance, and formalize each as a verifiable condition drawing on monetary economics (Fisher, 1911; Sidrauski, 1967), network effects theory (Metcalfe, 2013), commons governance theory (Ostrom, 1990), and recent token engineering literature. We then apply this diagnostic framework to three real community token economy projects currently at design stage, demonstrating how formal sustainability assessment can be conducted using only design-level parameters, token type, emission rules, burn mechanics, participant count, and actor structure, without requiring longitudinal transaction data.

This proposal is grounded in the work of the Comunità e Territori Digitali (Digital Communities and Territories) [1] research group, which focuses on the application of blockchain technology to collaborative

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[1] *Territori e Comunità Digitali* (Digital Communities and Territories) is a multidisciplinary research group within the Social

economies in local communities. For several years, the research has been carried out through action research, which combines literature and principles analysis with fieldwork through various projects based in local communities. Much of the effort has been devoted to the design and development of collaborative tools and the testing of the web app used to implement the projects. The aim of this paper is to provide an initial assessment of the token economies designed and to offer a diagnostic framework for evaluating their sustainability over time.

The three cases examined represent distinct examples of community token design: a volunteer reward system in which tokens flow one-directionally from a cooperative to participants who redeem them against local businesses and associations coupons; a peer-to-peer object lending system in which tokens allow the objects represented by NFT in the web app; and a food redistribution system in which token emission is anchored to physical quantities of surplus food.

Together, these cases provide a comparative basis for identifying both common structural vulnerabilities and case-specific risk profiles.

Our contribution is twofold. First, we provide the first formalization of sustainability conditions specifically adapted to community-scale token economies, distinguishing them from large-protocol tokenomics. Second, we demonstrate the empirical applicability of this framework through comparative case analysis, producing risk profiles for each system.

The paper is structured as follow: Section 2 contains the background of token economy design in collaborative and local communities context. Section 3 present the diagnostic framework through the formalization of six failure modes. Section 4 describes the three case studies and the risk assesment based on the mentioned framework. Section 5 discuss the findings and offer a comparison between the cases. Section 6 concludes.

2. Background

2.1. From Cryptoeconomics to Token Economies

The conceptual landscape surrounding blockchain-based economic systems is marked by significant terminological overlap and ambiguity [1]. Cryptoeconomics, the earlier of the foundational concepts, was initially defined as the design of systems using cryptography to verify past events and economic incentives to shape future behaviours [2], and has since been understood as encompassing interconnected layers of computation, financial exchange, and off-chain socioeconomic networks [3]. Tokenomics emerged as an evolution of this framework, expanding its scope to include the study of token issuance, consensus mechanisms, and broader game-theoretic incentive structures, with its foundations largely rooted in neoclassical economics [4]. Within this landscape, three related but distinct concepts warrant clarification. The token economy refers to systems designed to reward users based on their contributions, leveraging smart contracts to facilitate and incentivise transactions and platform participation. Tokenomics concerns the study of the supply, demand, distribution, and valuation of cryptocurrencies, including the mechanisms governing their issuance and utility. Tokenization, by contrast, denotes the process of converting real-world assets or rights into digital tokens on a distributed ledger, enabling the transfer of ownership without traditional intermediaries. Existing classification frameworks Lesche et al. [5], Pierluigi Freni [6], Tan [7] have sought to impose analytical order on this diversity, distinguishing token functions such as payment, utility, security, and governance, and classifying tokens along dimensions including fungibility, transferability, and regulatory status. However, a common limitation persists across these contributions: frameworks remain predominantly descriptive, offering taxonomies rather than formal sustainability criteria applicable at the design stage, and largely reflecting orthodox

Computing area of the Computer Science Department at the University of Turin. Combining a technical-scientific orientation with a socio-humanistic perspective, the group places the human being at the centre of its research, aiming to strengthen the relationship between physical and digital realities and to put technology at the service of contemporary challenges such as digital literacy, active citizenship, sustainability, and cooperation. The group develops civic digital technologies, innovative platforms for collaboration and participation that support community action and social economy business models, including FirstLife (https://firstlife.org/) and CommonsHood (https://commonshood.eu/).

economic thinking, leaving significant gaps in the understanding of value-driven token economies and their potential beyond simple transactional functions [8].

2.2. Monetary Economics and Token Supply Dynamics

The analytical foundation for evaluating token supply dynamics is Fisher's [9] equation of exchange, derived from the observation that total money paid for goods must equal the total value of goods exchanged in any community over a given period:

$$MV = \sum pQ \tag{1}$$

where M is the quantity of money in circulation, V is its velocity of circulation, and $\sum pQ$ is the sum of all prices multiplied by their respective quantities exchanged. In aggregate notation, written as $MV = PQ$, the price level varies directly with M and V, and inversely with transaction volume Q the core of the quantity theory of money [9]. Applied to token economies, this equation provides an internal consistency check: excess token supply relative to transaction demand manifests as internal inflation, eroding purchasing power and undermining participation incentives. Since community tokens have no external price anchor, P is entirely endogenous and $MV = PQ$ functions as a consistency condition rather than an absolute inflation measure.

A critical limitation of the aggregate equation is that it provides no insight into how individual agents contribute to velocity. Campajola et al. [10] address this by introducing *MicroVelocity*: factoring aggregate velocity into agent-specific contributions based on the distribution of token holding times τ. Defining $p_i(\tau)$ as the holding time distribution of agent i and M_i as their token balance, aggregate velocity decomposes as:

$$V = \sum_{i=1}^{N} \frac{M_i}{M} \int_0^\infty \frac{1}{\tau} p_i(\tau)\, d\tau \tag{2}$$

This formulation shows that in heterogeneous populations, individual behavioral differences, specifically, whether agents hold or spend tokens immediately, directly shape aggregate dynamics [10]. For this paper, MicroVelocity serves two functions: it allows velocity to be estimated at design stage from token design parameters alone, and it formally connects individual agent behavior to the system-level failure modes developed in Section 3. Sidrauski [11] provides the dynamic complement, establishing that sustainable monetary expansion must remain proportional to real economic growth, a condition operationalized in the burn/emission imbalance failure mode.

2.3. Commons Governance and Collective Action

The third theoretical pillar is Ostrom's theory of commons governance [12]. Against the assumption that common-pool resources inevitably collapse through overexploitation, usually referred to as the tragedy of the commons [13], Ostrom demonstrated that communities can design sustainable collective institutions provided certain design principles are met.

Three of Ostrom's eight principles are directly relevant here. *Congruence* requires that benefits be proportional to contributions, directly motivating the contribution balance condition in Section 3. *Collective choice arrangements* require that affected participants be able to modify governing rules, underpinning the governance capture failure mode. *Monitoring* requires that both resources and behaviors be observed by accountable parties; blockchain transparency partially satisfies this condition, though monitoring of off-chain behavior and contribution quality remains a design challenge in community token systems.

Ostrom [14] further addresses the behavioral conditions for collective action, identifying participation thresholds and contribution dynamics that map closely onto token economy participation questions. The key insight for this research is that sustainable commons do not rely on purely financial incentives: social norms, reciprocity, and community identity play central roles in sustaining contribution behavior.

This observation suggests that community token economies operating in collaborative or solidarity contexts may exhibit systematically different behavioral dynamics than those predicted by orthodox tokenomic models, including longer token holding times, higher contribution rates, and greater tolerance for distributional asymmetries, provided that Ostrom's governance conditions are met.

These qualitative principles have been progressively formalized through evolutionary game theory and spatial dynamics. Replicator dynamics demonstrates that without institutional constraints, defection is the unique evolutionary attractor in well-mixed populations [15], providing the formal baseline against which Ostrom's design principles operate. Spatial game theory further shows that locally structured interaction, characteristic of a local community economies, can sustain cooperative clusters even where well-mixed models predict collapse [15]. Behavioral economics additionally identifies loss aversion and hyperbolic discounting as systematic deviations from rational-agent assumptions that compound sustainability risk in practice [16, 15]. The implications of these extensions for the diagnostic framework proposed in this paper are discussed in Section 5.

3. A Diagnostic Framework for Community Token Economy Sustainability

This section presents a diagnostic framework comprising six formally grounded failure modes that threaten the sustainability of community token economies. For each failure mode, we provide a conceptual definition, a formal condition derived from the theoretical foundations established in Section 2, and the design parameters required to evaluate it. The framework is explicitly designed for application at *design stage*, using only parameters available from system documentation prior to deployment.

We define a community token economy as *sustainable* if it satisfies simultaneously four aggregate conditions:

- **Supply stability:** token supply remains bounded relative to real transaction demand
- **Participation viability:** participation remains above a critical threshold sufficient to sustain liquid exchange
- **Incentive balance:** contributions and benefits remain proportionally balanced, and tokens retain sufficient value to reward participation
- **Governance integrity:** governance authority is sufficiently distributed to reflect community self-determination

The six failure modes operationalize the conditions under which one or more of these properties breaks down. Supply stability and incentive balance each admit two structurally distinct failure pathways, giving rise to six failure modes organized across four sustainability dimensions:

- **Supply stability** → Failure Mode 1 (token oversupply) and Failure Mode 3 (burn/emission imbalance)
- **Incentive balance** → Failure Mode 2 (velocity trap) and Failure Mode 4 (free-rider collapse)
- **Participation viability** → Failure Mode 5 (insufficient critical mass)
- **Governance integrity** → Failure Mode 6 (governance capture)

Failure Modes 1, 2, and 3 form a reinforcing supply-side triad: velocity trap amplifies inflationary pressure from oversupply, and the absence of a burn mechanism removes the only structural corrective. The framework should therefore be applied holistically rather than as a checklist of isolated conditions.

3.1. Failure Mode 1: Token Oversupply and Inflation Spiral

Definition. Token oversupply occurs when the rate of token emission persistently exceeds the growth of real economic activity within the community, resulting in a decline in the internal usability of the token. As token value erodes, participation incentives weaken, potentially triggering a self-reinforcing spiral of declining engagement and further devaluation.

Sustainability threshold. Starting from Fisher's equation of exchange [9]:

$$MV = PQ \tag{3}$$

taking logarithms and differentiating with respect to time yields the continuous-time inflation rate:

$$\pi := \frac{\dot{P}}{P} = \frac{\dot{M}}{M} + \frac{\dot{V}}{V} - \frac{\dot{Q}}{Q} \tag{4}$$

Inflation occurs when $\pi > 0$, i.e. when:

$$\frac{\dot{M}}{M} + \frac{\dot{V}}{V} > \frac{\dot{Q}}{Q} \tag{5}$$

In discrete time, the oversupply condition is:

$$\frac{\Delta(MV)}{MV} > \frac{\Delta Q}{Q} \tag{6}$$

A critical implication of Equation 4 is that inflationary pressure arises from the product MV, not from token supply M alone. Even if token issuance is controlled, if agents rush to spend, driving velocity V upward, the system behaves as if it is oversupplied. Formally, substituting the MicroVelocity decomposition of Campajola et al. [10]:

$$P = \frac{M \cdot \sum_i \frac{M_i}{M} \int_0^\infty \frac{1}{\tau} p_i(\tau)\, d\tau}{Q} \tag{7}$$

this shows that inflationary pressure is amplified when short holding times compound excess emission, linking Failure Mode 1 directly to Failure Mode 2. The three paths to inflation are:

- **Pure oversupply:** $\dot{M}/M > \dot{Q}/Q$ with $\dot{V}/V = 0$
- **Pure velocity trap:** $\dot{V}/V > 0$ with $\dot{M}/M = 0$ (formalized in Failure Mode 2)
- **Combined failure:** both simultaneously, producing maximally amplified inflationary pressure

At design stage, Equation 6 is evaluated by comparing the planned emission schedule against expected transaction volume, estimated from participant count and intended activity frequency. The consistent token supply at a given velocity is:

$$M_{consistent} = \frac{PQ}{V} \tag{8}$$

If planned emission significantly exceeds $M_{consistent}$, the system exhibits structural oversupply at launch.

Parameters required. Emission rate $E(t)$; expected transaction volume Q; participant count N; average activity frequency per participant; velocity estimate V (see Failure Mode 2).

3.2. Failure Mode 2: Token Velocity Trap

Definition. As established in Failure Mode 1, inflationary pressure depends not on token supply alone but on the product MV. The velocity trap failure mode identifies the behavioral mechanism through which high circulation speed amplifies inflationary pressure: when tokens have no structural reason to be held rational agents spend immediately upon receipt, driving velocity upward and compounding any existing supply imbalance. In community token economies, this dynamic is particularly consequential because the perceived reward for participation erodes as token purchasing power declines.

Sustainability threshold. From the MicroVelocity framework of Campajola et al. [10], aggregate system velocity is:

$$V = \sum_{i=1}^{N} \frac{M_i}{M} \int_0^\infty \frac{1}{\tau} p_i(\tau)\, d\tau \tag{9}$$

where $p_i(\tau)$ is the holding time distribution of agent i and M_i is their token balance. Defining the wealth-weighted average holding time:

$$\bar{\tau} = \sum_{i=1}^{N} \frac{M_i}{M} \cdot \mathbb{E}_i[\tau] \tag{10}$$

the approximation $V \approx 1/\bar{\tau}$ is commonly used but introduces a systematic underestimation. Since $f(x) = 1/x$ is convex, Jensen's inequality [17] gives:

$$V = \mathbb{E}\left[\frac{1}{\tau}\right] \geq \frac{1}{\mathbb{E}[\tau]} = \frac{1}{\bar{\tau}} \tag{11}$$

Therefore $1/\bar{\tau}$ is a *lower bound* on velocity, not an equality: a small subgroup of fast-spending agents can drive system velocity significantly above what the average holding time suggests. For design-stage assessment, this conservative lower bound nonetheless provides a useful diagnostic: a velocity trap is indicated when:

$$\bar{\tau} \to 1 \tag{12}$$

meaning the wealth-weighted average holding time approaches a single period, consistent with immediate token spending upon receipt. Since $1/\bar{\tau}$ underestimates true velocity, a system exhibiting $\bar{\tau} \approx 1$ at design stage is at minimum at the velocity trap boundary, and likely beyond it once agent heterogeneity is accounted for. The full implications of behavioral heterogeneity for velocity estimation, including the precise condition $V > V^*$, are discussed in Section 5.

At design stage, velocity trap risk is assessed by examining whether the token design includes any structural mechanism that rewards or necessitates holding: staking, governance rights, time-locked rewards, or tiered pricing that requires multi-period token accumulation.

Parameters required. Token utility structure; existence of holding incentives; expected holding time $\mathbb{E}_i[\tau]$ by agent type; token balance distribution M_i/M across agent types.

3.3. Failure Mode 3: Burn/Emission Imbalance

Definition. Failure Mode 3 identifies the structural absence of the corrective mechanism that would otherwise stabilize supply over time. In the absence of a burn mechanism, total token supply grows monotonically with each emission event, and the inflationary pressures identified in Modes 1 and 2 cannot self-correct. Even when burn mechanisms exist, their effectiveness depends critically on whether the burn rate responds to actual system usage rather than following a fixed schedule independent of transaction volume.

Sustainability threshold. Let $E(t)$ denote the token emission rate and $B(t)$ the burn rate at time t. Net supply change is:

$$\dot{M} = E(t) - B(t) \tag{13}$$

The sustainability condition, derived from Sidrauski [11], requires that net supply growth does not exceed the growth rate of community economic activity:

$$E(t) - B(t) \leq g(t) \cdot M(t) \tag{14}$$

where $g(t)$ is the participant or transaction growth rate at time t. For a stable or slowly growing community ($g(t) \approx 0$), this reduces to:

$$E(t) \approx B(t) \tag{15}$$

When $B(t) = 0$ — as in systems with no burn mechanism — the condition is violated for any positive emission rate. The burn coverage ratio:

$$\rho = \frac{B(t)}{E(t)} \tag{16}$$

provides a simple design-stage diagnostic: $\rho \geq 1$ is required for supply stability under zero growth. Values of $\rho < 1$ indicate structural long-run inflationary pressure that compounds the dynamics identified in Failure Modes 1 and 2.

It is important to note that Equation 14 is a necessary but not fully sufficient condition: from Equation 4, a complete treatment must also account for velocity dynamics, since increasing spending speed requires supply tightening even when transaction volume stays constant. The full velocity-dependent condition and an adaptive supply controller are developed in Section 5.

For burn mechanisms to be effective, the burn trigger must be *demand-driven* — $B(t)$ should be a function of $Q(t)$ rather than a fixed rule independent of actual usage, as a rule-driven burn creates no structural link between token destruction and real economic value.

Parameters required. Emission rate $E(t)$; burn mechanism existence and trigger type; whether burn is demand-driven or rule-driven; burn coverage ratio ρ; participant or transaction growth rate $g(t)$.

3.4. Failure Mode 4: Free-Rider and Contribution Collapse

Definition. In community token economies, system value depends on participants contributing goods, services, or labor. Free-riding, consuming system benefits without contributing, reduces the supply of available goods and services, degrading system value. If the token design provides no structural incentive for contribution, or if the reward for contributing is insufficient relative to effort, the contributor rate will decline and the system may tip into a collapse cascade where fewer contributors means less value means even fewer contributors.

Formal condition. Let $\phi \in [0, 1]$ denote the fraction of participants who actively contribute at any given period, N the total participant count, d the average demand per participant per period, and K the number of distinct offer types. Drawing on Ostrom's [1990] proportionality principle, the minimum viable contribution condition is:

$$\phi \cdot N \geq \frac{d \cdot N}{K} \quad \Longleftrightarrow \quad \phi \geq \frac{d}{K} \tag{17}$$

This states that the active contributor rate must be at least as large as average demand per person divided by offer variety. The formal condition for institutional stability under monitoring, derived from Li et al. [15], further requires:

$$\gamma S > T - R \tag{18}$$

where γ is the probability of detection of free-riding, S is the sanction magnitude, T is the temptation payoff from defecting, and R is the reward from mutual cooperation. This formalizes Ostrom's graduated sanctions principle: high monitoring frequency γ, which blockchain transparency can partially provide, allows for lower sanctions S while maintaining incentive compatibility.

Parameters required. Token earning mechanism; contribution verification mechanism; offer variety K; average demand d per participant; monitoring capacity γ; sanction structure S.

3.5. Failure Mode 5: Insufficient Critical Mass

Definition. Token economies require a minimum viable number of active participants to generate sufficient exchange transactions for tokens to be useful and liquid. Below this threshold, the system is technically operational but socially inert: tokens accumulate without being spent, and participants perceive no value in earning them.

Formal condition. For a community of N participants with K distinct offer types, the expected number of successful matches per period, derived from the double coincidence of wants framework, is:

$$\mathbb{E}[\text{matches}] = \frac{N(N-1)}{2} \cdot \frac{1}{K} \tag{19}$$

The system is above the critical mass threshold if:

$$\frac{N(N-1)}{2K} \geq \delta \tag{20}$$

where $\delta = d \cdot N$ is the minimum number of transactions per period required to maintain token liquidity. Solving for the minimum participant count:

$$N \geq 2Kd + 1 \tag{21}$$

This provides a design-stage estimate of the minimum community size required to sustain liquid token exchange, as a function of offer variety K and average demand d.[2]. It is important to note that this condition assumes a well-mixed population; in spatially structured communities where interaction is locally bounded, cooperative clusters may sustain exchange below this threshold through spatial reciprocity [15], making Equation 21 a conservative upper bound on critical mass risk.

Parameters required. Participant count N; offer variety K; average demand per participant d; interaction topology (well-mixed vs. spatially structured).

3.6. Failure Mode 6: Governance Capture

Definition. Governance capture occurs when decision-making authority over token economy parameters, emission rates, burn rules, eligibility criteria, exchange ratios, concentrates in a small subset of participants or in a central authority, undermining the community character of the system. In Ostrom's [12] terms, governance capture violates the collective choice arrangements and recognition principles, creating structural dependency on the continued goodwill and competence of the controlling entity.

Formal condition. At design stage, governance capture is assessed through the *governance centralization index* $\Gamma \in [0, 1]$, defined as the proportion of governance decisions that can be taken unilaterally by a single actor or small group:

$$\Gamma = \frac{|\text{unilateral decisions}|}{|\text{total governance decisions}|} \tag{22}$$

A system with $\Gamma = 1$ is fully centralized; $\Gamma = 0$ requires full consensus. We propose a sustainability threshold of $\Gamma \leq 0.5$ as a minimum condition for meaningful distributed governance. The token distribution Gini coefficient G over participant wallets provides a complementary metric: rising G over time signals progressive concentration of token-based influence, warranting monitoring as a secondary governance indicator.

[2]This condition is derived from the double coincidence of wants framework [18, 19] by setting $\mathbb{E}[\text{matches}] \geq \delta = dN$ and solving for N, assuming a well-mixed population with uniform demand distribution. Market thickness theory [20] confirms that matching probability increases significantly with participant count, supporting the use of this threshold as a design-stage diagnostic. In spatially structured communities, spatial reciprocity [15] may sustain exchange below this threshold, making the condition a conservative upper bound on critical mass risk.

Parameters required. Governance rule structure; token creation and modification rights; actor roles and decision authorities; initial token distribution.

3.7. Framework Summary

Table 1 summarizes the six failure modes, their formal conditions, and the required design parameters, organized by sustainability dimension.

Table 1
Diagnostic Framework: Six Failure Modes for Community Token Economies

#	Failure Mode	Formal Condition	Condition	Key Parameters
1	Oversupply / Inflation	$\Delta(MV)/MV > \Delta Q/Q$	Supply stability	Emission rate, Q, $\bar{\tau}$
2	Velocity Trap	$\bar{\tau} \to 1$	Supply stability	Holding incentives, $\mathbb{E}_i[\tau]$
3	Burn/Emission Imbalance	$E(t) - B(t) > g(t) \cdot M(t)$	Supply stability	Burn mechanism, $g(t)$
4	Free-Rider Collapse	$\phi < d/K; \gamma S \leq T - R$	Incentive balance	ϕ, K, d, γ, S
5	Insufficient Critical Mass	$N < 2Kd + 1$	Participation viability	N, K, d
6	Governance Capture	$\Gamma > 0.5$	Governance integrity	Decision rights, G

The six conditions are explicitly presented as *minimum thresholds* rather than optimal values: satisfying all six is a necessary but not sufficient guarantee of sustainability. The conditions interact in ways that require holistic rather than checklist-style assessment.

4. Case Studies and Application of the Framework

The diagnostic framework is applied to three community token economy projects developed within the *Digital Communities and Territories* research group through an action research approach combining theoretical analysis with fieldwork in local communities. All three are built on the CommonsHood wallet application and are currently at design or early experimentation stage. Due to space constraints, this section presents condensed risk profiles for each case rather than a full step-by-step application of the diagnostic framework.

Table 2 summarizes the key tokenomic parameters across cases.

4.1. Case 1: NLAB4CIT — Volunteer Reward System

The NLAB4CIT pilot, developed through a collaboration between the Municipality of Collegno and the University of Turin, implements a blockchain-based token reward system for civic participation [21]. Volunteers earn tokens through association activities (10 tokens/week full participation; 5 tokens/session spot engagement) and redeem them for coupons at eight local merchants and socio-cultural providers, with coupon values of 5, 10, or 15 tokens (equal distribution, $\bar{P} = 10$ tokens). Tokens terminate at merchant wallets without recirculation.

4.2. Case 2: Library of Things — CORPUS Project

The Library of Things, developed within the Turin Urban Living Lab as part of the CORPUS project, is a peer-to-peer object lending platform where each item is represented by an NFT [22]. Users borrow items at 1 token/day, with tokens held in escrow during the loan period and released to the lender

Table 2
Key Design Parameters Across Three Cases

Parameter	NLAB	CORPUS	FoodReborn
Participants N	≈ 100	Community-scale	Beneficiaries
Token name	Reward Token	COSO	Food Token
Token type	ERC-20 fungible	ERC-20 + NFT	ERC-20 fungible
Emission basis	Participation events	Platform actions	Food weight × price
Emission rate E	≈750 tokens/week	Variable (rewards)	Variable (food collected)
Coupon/exchange	8 merchants, 5 coupons each; 5/10/15 tokens	1 token/day borrowing	Meals at social restaurants
Burn mechanism	None	None	Coupon burn at restaurant
Token recirculation	None (terminal)	Partial (lender receives)	None (terminal)
Governance	Fully centralized	Developer-controlled	Centralized (Project Manager)
Value anchor	Behavioral events	Time-based pricing	kg × market price

upon return. Multiple reward mechanisms incentivize prosocial behavior: membership (10 tokens), community (2 tokens/upload), trust (1 token/ completed loan), and care (2 tokens/good return). Physical handover is required for all exchanges.

4.3. Case 3: FoodReborn — Food Poverty Redistribution

The FoodReborn pilot tokenizes surplus food collected from Turin's CAAT wholesale market using a real-world value anchor: 1 kg × average daily CAAT price = token value (e.g. 100 kg × €2.00 = 200 tokens). Beneficiaries receive tokens and redeem them for coupons at social restaurants visible on a geolocated map. Coupons are burned upon redemption at the restaurant, closing the exchange cycle and preventing token reuse.

4.4. Diagnostic Results and Cross-Case Comparison

Table 3 presents the structured risk profiles for all three cases.

The comparative analysis reveals four structural patterns that characterize community token economies as a class, independent of their specific social purpose or architecture.

Universal absence of token-level burn mechanisms. None of the three systems incorporates a burn mechanism at the token layer ($\rho = 0$ in all cases), meaning all exhibit monotonically growing token supply regardless of actual system usage. Case 3 is the closest to an exception: its coupon burn at the restaurant level constitutes an effective terminal mechanism for coupons, but the underlying token supply remains uncorrected. This is the most actionable finding of the cross-case analysis: all three systems would benefit from introducing a demand-driven token burn triggered at the point of service redemption, linking token destruction directly to real exchange activity.

Inverted risk profiles across supply and matching. The three cases exhibit markedly different risk concentrations reflecting their distinct exchange architectures. Case 1 faces the most severe supply-side risk, all three supply failure modes (FM1, FM2, FM3) are simultaneously violated, forming the reinforcing

Table 3
Diagnostic Risk Profiles Across Three Cases

Failure Mode	Case 1	Case 2	Case 3
FM1: Oversupply	**Critical** $E/M_{cons} \approx 2.5$	**Low** $E/M_{cons} \approx 0.10$	**Low–Med.** Value-anchored; price volatile
FM2: Velocity trap	**High** $\bar{\tau} \approx 1.5$; no holding incentive	**Low** $\bar{\tau} \approx 7$ days; escrow dampens V	**Med.** High redemption pressure
FM3: Burn/emission	**Critical** $\rho = 0$	**Low–Med.** $\rho = 0$; self-balancing flow	**Low–Med.** Coupon burn; token layer uncorrected
FM4: Free-rider	**Low** (volunteers) **High** (merchants)	**Low** Self-correcting by design	**Med.** Institutional actors unincentivized
FM5: Critical mass	**Satisfied** $N = 100 \geq N^* = 9$	**High** Item-specific matching	**Low** Fixed locations; logistics threshold
FM6: Governance	**Centralized** $\Gamma = 1.0$; no roadmap	**Centralized** $\Gamma \approx 0.8$; no roadmap	**Centralized** $\Gamma \approx 1.0$; no roadmap

triad, with an oversupply ratio of $E/M_{cons} \approx 2.5$, but has the simplest matching structure with coupons available at fixed merchant locations. Case 2 presents the inverse profile: a structural undersupply ratio of $E/M_{cons} \approx 0.10$, with physical lending cycles and escrow mechanisms naturally dampening velocity to $V \approx 0.14$ per period, but facing the hardest matching problem due to item-specific double coincidence of wants. The quantitative contrast between the two cases demonstrates that token standard alone is not a reliable predictor of supply sustainability; the exchange mechanism and holding time structure are the decisive parameters.

Case 3 occupies an intermediate position, with a real-world value anchor partially constraining emission and a coupon burn operating one layer above the token supply. These inverted profiles reveal a fundamental design tension: architectural features that improve supply stability, physical anchoring, escrow, fixed locations, tend to increase matching difficulty, while features that facilitate matching tend to increase velocity and supply instability. This trade-off cannot be optimized simultaneously without deliberate architectural choices, suggesting that different community contexts require fundamentally different tokenomic designs rather than a single universal model.

Governance centralization as a bootstrapping necessity. All three systems exhibit significant governance centralization ($\Gamma \geq 0.8$) at their current stage, reflecting the practical necessity of trusted intermediaries to facilitate community onboarding with complex blockchain technology. This is consistent with the bootstrapping phase of commons institutions identified by Ostrom [12], where a founding authority often precedes self-governance. Centralization at launch is therefore not a design failure but an expected feature of early-stage community token economies where technical literacy and institutional capacity are still developing.

The sustainability risk flagged by Failure Mode 6 is more precisely the *absence of a planned transition pathway* toward distributed governance as community capacity matures. None of the three cases describes explicit mechanisms for progressively delegating governance authority to participants which constitutes the actual design gap.

5. Discussion

5.1. Limitations of the Diagnostic Framework

The diagnostic framework presented in this paper is explicitly designed as a first-order approximation, and several limitations must be acknowledged in interpreting its results.

The most fundamental is the *well-mixed population assumption* underlying the critical mass condition (Failure Mode 5). The matching formula $N \geq 2Kd + 1$ assumes that any participant can interact with any other with equal probability, a condition that does not hold in geographically bounded, socially structured community economies. Spatial game theory [23] demonstrates that in locally structured populations, cooperators can self-organize into clusters sustained by spatial reciprocity, potentially maintaining viable exchange below the theoretical critical mass threshold. The formula therefore provides a conservative upper bound on participation risk rather than a precise prediction: real community token economies embedded in dense social networks may function effectively at participant counts below N^*, particularly when interaction is anchored to fixed physical locations as in Cases 1 and 3.

A second limitation concerns the *rational agent baseline* of the framework. The sustainability conditions derived from Fisher and Sidrauski assume agents who maximize utility consistently over time. Behavioral economics identifies two systematic deviations that are particularly relevant for community token economies. Prospect theory [16] demonstrates that agents evaluate outcomes as gains and losses relative to a reference point, with losses weighted approximately 2.25 times more heavily than equivalent gains. In token economy contexts, burn mechanisms and sanctions are experienced as losses relative to current token balances, facing stronger psychological resistance than rational models predict. Hyperbolic discounting [15] further suggests that participation commitment may decay over time as present bias causes agents to undervalue future token utility relative to immediate alternatives, a dynamic particularly relevant for Case 1 where volunteers must accumulate tokens over multiple weeks to access higher-tier coupons. Together these behavioral mechanisms suggest that the sustainability thresholds derived in Section 3 represent conservative lower bounds on real-world risk.

A third limitation concerns the *velocity approximation*. As established in Section 3, the formula $V \approx 1/\bar{\tau}$ systematically underestimates true velocity due to Jensen's inequality:

$$V = \mathbb{E}\left[\frac{1}{\tau}\right] \geq \frac{1}{\mathbb{E}[\tau]} = \frac{1}{\bar{\tau}} \tag{23}$$

A small subgroup of fast-spending agents can drive system velocity significantly above what the average holding time suggests, pushing the system toward inflationary conditions that population-average estimates do not predict. The full velocity-dependent sustainability condition on emission, derived from Equation 4, is:

$$\frac{\dot{M}}{M} \leq \frac{\dot{Q}}{Q} - \frac{\dot{V}}{V} \tag{24}$$

This formulation makes explicit that if agents begin spending faster, driving $\dot{V}/V > 0$, supply must tighten correspondingly even if transaction volume stays constant. A fixed emission schedule is therefore structurally insufficient in systems prone to behavioral heterogeneity in spending habits. Monitoring velocity dynamics in real time and adjusting emission accordingly is a necessary but currently absent feature in all three cases examined.

Finally, the *matching model* underlying Failure Mode 5 deserves critical scrutiny. The matching probability $P(\text{match}) = 1/K$ assumes barter-like exchange where two agents must simultaneously have what the other wants. This assumption is partially inconsistent with the token economy framework itself, since tokens are precisely designed to decouple supply and demand and eliminate the double coincidence problem. The $1/K$ assumption is most defensible for the *redemption* side of the exchange, where participants must find a specific coupon, item, or service they actually want, rather than for token circulation, which is already decoupled. This distinction is particularly important for Case 2,

where borrowers must find a specific item, and less critical for Cases 1 and 3, where coupons and meals are more fungible. A network-based matching model [24] that captures community structure and interaction topology would provide a more realistic critical mass estimate, and is identified as a priority extension of the framework.

5.2. Rethinking Velocity in Community Currency Contexts

The velocity trap formalization in Section 3 requires important qualification when applied to community and complementary currency systems. The standard monetary economics framework from which the velocity analysis derives treats high velocity as pathological, a signal that agents are offloading currency as fast as possible, indicating low confidence in its value. This logic is appropriate for national currencies designed as stores of value, but it conflicts with the theoretical foundations of complementary currency design.

The complementary currency tradition, originating with Gesell's [25] concept of *Freigeld*, money designed to lose value when hoarded through a demurrage fee, and developed by Lietaer and Dunne [26], argues the opposite: in community economies suffering from insufficient exchange, high velocity is a sign of a healthy, active system. Community currencies are explicitly designed to circulate rapidly rather than accumulate, redistributing value locally and stimulating exchange that mainstream currencies suppress through scarcity and hoarding incentives. From this perspective, $\bar{\tau} \to 1$ is the *design goal*, not a failure mode.

Reconciling these two frameworks requires recognizing that the velocity trap is not simply a condition on V but on the *relationship between V and absorptive capacity Q*. High velocity is only pathological when it outpaces real exchange activity, when tokens circulate faster than goods and services are actually being exchanged, generating inflationary pressure. Formally, the revised velocity trap condition is:

$$V > V^* := \frac{PQ}{M} \tag{25}$$

where V^* is the maximum velocity consistent with stable token value at the current supply and transaction volume. A system with high velocity but equally high transaction volume, meaning every token that circulates quickly corresponds to a real exchange event, satisfies $V \leq V^*$ and is healthy by this condition regardless of how fast tokens move. The problem in Case 1 is not that volunteers spend tokens quickly, but that they spend them faster than the coupon market can absorb, leaving excess tokens with nowhere to go.

This reframing also resolves the apparent paradox in Case 2. The Library of Things has low velocity by design ($V \approx 0.14$ per period), which looks healthy from an orthodox monetary perspective, but this low velocity is a structural constraint imposed by the physical lending cycle rather than a deliberate design choice. A system with higher velocity and proportionally higher transaction volume would be equally sustainable by the revised condition, and potentially more vibrant as a community exchange platform.

A further consideration concerns the K parameter that appears in both the free-rider condition ($\phi \geq d/K$) and the critical mass condition ($N \geq 2Kd + 1$). These two conditions reveal an important and underappreciated design tension: increasing offer variety K makes the free-rider condition easier to satisfy, demand is spread more thinly across types, so a lower contributor rate suffices, but simultaneously makes the critical mass condition harder to satisfy, since finding a compatible trading partner becomes less probable. There is therefore an optimal K that balances contribution ease against matching difficulty, and this optimum depends on the specific social context and participant base of each community token economy. Identifying this optimum analytically is a natural extension of the present framework.

5.3. Towards Agent-Based Modeling

This paper is explicitly framed as a preliminary contribution to a broader research agenda whose short-term goal is the development of a formal agent-based model (ABM) of community token economy dynamics. The diagnostic framework developed here serves a dual purpose in this agenda.

First, it provides the *parameter space* for simulation. The six minimum threshold conditions define the axes along which agent behavior and system dynamics will be varied in the computational model: emission rate relative to absorptive capacity, velocity relative to V^*, burn coverage ratio, contributor rate, participant count, and governance centralization. The case study data provides empirically grounded initial conditions for each axis, ensuring that simulations begin from realistic rather than arbitrary starting points.

Second, it provides the *failure mode vocabulary* for interpreting simulation outputs. Rather than simply observing whether a simulated community token economy survives or collapses, the agent-based model will be designed to identify which specific failure mode drives observed outcomes. This allows simulation results to be mapped back onto the diagnostic framework, producing actionable design insights rather than opaque emergent behavior.

The transition from the present analytical framework to an agent-based model requires formalizing individual agent decision rules (contribution versus free-riding, holding versus spending, participation versus exit) and specifying how these rules interact with the token economy parameters identified here. Agent-based modeling is uniquely suited to this task for three reasons. It naturally accommodates the behavioral heterogeneity identified as a limitation of the current framework, allowing fast-spending and slow-spending agent types to coexist and interact. It can incorporate spatial structure, relaxing the well-mixed population assumption and capturing the clustering dynamics identified by spatial game theory [15] as a potential moderator of free-rider collapse and critical mass risk. And it generates emergent system-level dynamics from micro-level behavioral rules, allowing the reinforcing triad of supply-side failure modes to be studied as a dynamic process rather than a static condition.

The replicator dynamics framework [15] provides a natural bridge between the analytical conditions derived here and the agent update rules needed for simulation: the free-rider condition $\phi \geq d/K$ maps onto the cooperator frequency threshold below which defection becomes the evolutionary attractor, and the monitoring condition $\gamma S > T - R$ maps onto the institutional parameters that shift the fitness landscape toward cooperation. The present paper thus constitutes both a standalone empirical contribution and the analytical foundation for a forthcoming simulation-based investigation of community token economy sustainability.

6. Conclusion and Future Works

This paper has addressed a fundamental open question in the design of community token economies: what are the minimum parameter conditions that must be satisfied prior to deployment to prevent structural failure? We have proposed a diagnostic framework of six formally grounded failure modes (token oversupply, velocity trap, burn/emission imbalance, free-rider collapse, insufficient critical mass, and governance capture) each associated with a minimum threshold condition derived from monetary economics, commons governance theory, token engineering literature, and matching theory. The framework has been applied to three real community token economy projects at design stage, producing structured risk profiles and cross-case comparative findings.

The analysis reveals three principal contributions. First, the framework demonstrates that design-stage sustainability assessment is feasible using only parameters available from system documentation prior to deployment without requiring longitudinal transaction data. This makes it applicable at the moment when design corrections are most feasible and least costly.

Second, the comparative analysis surfaces systematic structural vulnerabilities common across community token economies as a class: the universal absence of token-level burn mechanisms, the governance centralization inherent to the bootstrapping phase, and the fundamental design tension between supply stability and matching viability. The 25-fold difference in supply dynamics between Case 1 ($E/M_{cons} \approx 2.5$) and Case 2 ($E/M_{cons} \approx 0.10$), despite both using the ERC-20 standard, demonstrates that token standard alone is not a reliable predictor of sustainability; exchange mechanism and holding time structure are the decisive parameters.

Third, the paper identifies a previously underexplored design parameter, the emission basis, as

critical for supply stability. Case 3's food-weight anchor represents a design pattern in which token issuance is structurally bounded by real resource flows, avoiding the unconstrained supply growth that characterizes participation-event emission in Cases 1 and 2.

A natural next step, currently under development, is an automated *token economy verifier* that encapsulates the six conditions within a structured interface accessible to non-technical practitioners. Rather than relying on point estimates, the verifier employs parameter-box reasoning: the user declares a range for each uncertain parameter, and the tool determines whether each failure mode condition is *sound* (the safety predicate holds across the entire declared range), *fragile* (the specification straddles the failure threshold, with both safe and unsafe assignments possible), or *broken* (no assignment satisfies the condition, indicating that structural redesign is required). This approach surfaces cross-parameter coherence issues that point-estimate analysis cannot detect, such as the incompatibility between a high resilience requirement and the absence of a burn mechanism, and makes the framework operationalizable by the community managers, social cooperatives, and local associations for whom it was designed. The verifier will be paired with an agent-based simulation backend that projects the declared specification forward in time, enabling practitioners to observe how failure modes manifest dynamically before deployment.

Several limitations accompany these contributions. The framework assumes well-mixed populations, rational agents, and static parameters, all of which are relaxed in real community systems. The velocity approximation $V \approx 1/\bar{\tau}$ systematically underestimates true velocity due to behavioral heterogeneity among fast-spending agents. And the matching model underlying the critical mass condition is most appropriate for redemption-side matching rather than token circulation. These limitations are not incidental but productive: they define precisely the research agenda that the forthcoming agent-based modeling investigation will pursue, relaxing each assumption in turn to test the robustness of the diagnostic conditions under more realistic behavioral and structural parameters.

Community token economies represent an emerging and undertheorized domain at the intersection of blockchain technology, collaborative economics, and commons governance. The framework proposed here is a first step toward giving designers of these systems the formal tools they need to anticipate failure before it occurs, rather than diagnosing it after the fact. The cases examined suggest that the gap between tokenomic theory and community practice remains wide, but that it is open to systematic analysis and that bridging this gap represents both a theoretical challenge and a practical necessity for communities that depend on these systems.

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Declaration on Generative AI

During the preparation of this work, the authors used Claude (Anthropic, claude-sonnet-4-6) to support the following activities: Grammar and spelling check; mathematical formalization and formal verification of sustainability conditions; After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

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